Domain 2: Decision Intelligence

EZ SYSTEM

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Introduction

EZ_System is a smart e-commerce platform that turns messy business data into clear and useful insights for small and medium sellers. It combines sales, stock, and external economic data like CPI and PPI to help users understand their business better. Using both rule-based logic and AI, it give clear advice like pricing decisions, stock planning, and demand forecasts. This helps sellers make better decisions using data instead of guessing, improving profit in a changing market.

Problem Statement

Many e-commerce SMEs struggle with "data blindness," where they have plenty of sales records but no clear way to turn them into smart decisions. Traditional tools are often too complex or only show raw numbers, forcing sellers to rely on guesswork for pricing and inventory. This problem is made worse by a failure to account for external economic factors like inflation, leading to hidden losses, inefficient stock management, and reduced profit margins in a volatile market.

Objective

- To bridge raw business data and external economic indicators into a unified AI-driven system that delivers clear, automated, and explainable decision-making insights (pricing, inventory, and demand forecasts) for non-technical users.

Target User

E-commerce Sellers

Individuals or small teams operating on platforms like Shopee, TikTok Shop, or Lazada who need to manage high-velocity inventory and competitive pricing.

Small Business Entrepreneurs

Owners of independent online or physical stores who currently rely on intuition or basic spreadsheets and want to scale using data-driven insights.

Methodology

Stage	Technology	Objective
Data Layer	MySQL / Python (sqlalchemy)	Consolidate fragmented e-commerce data.
Logic Layer	Rule Engine	Detect critical business risks (Low stock, loss).
AI Layer	Z.AI GLM-5.1	Generate strategic advice & economic reasoning.
Interface Layer	Flutter	Present human-readable decision intelligence.

Description of the Interaction System

User Action → Data Sync → AI Reasoning → Actionable Advice

User Action

The interaction begins with the user providing structured business data through a user-friendly CRUD (Create, Read, Update, Delete) interface.

- User Input: Product details, costs, and sales figures.
- System Input: The backend automatically fetches external economic data (CPI/PPI) and merges it with the user’s internal records to create a "Holistic Business Snapshot."

Data Sync

Once data is submitted, the Sync Engine acts as the intermediary. It triggers a background process that aggregates fragmented tables into a unified analytical summary . This ensures the AI isn't just looking at random numbers but at a calculated "Business Trace" (for example: calculating conversion rates and profit margins automatically).

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AI Reasoning

The core interaction occurs via the Z.AI GLM Engine.

- The Request: The system sends the "Business Trace" to the AI using a structured prompting strategy.
- The Reasoning: The AI performs a "Trade-Off Analysis," weighing internal constraints (low stock) against external factors (high inflation).
- The Output: Instead of complex code, the AI returns human-readable JSON, which the system parses into Strategic Actions and Impact Summaries.

Actionable Advice

The final interaction happens on the Analytics Dashboard (Flutter Frontend).

- Strategic Cards: Recommendations are presented as high-impact cards (e.g. "Strategic Action", "Forecast" and “Trade-off).
- Interactive Trigger: Users can manually trigger a analysis using the "Activate AI Analysis" button, allowing them to see how their business health changes after they update prices or restock products.

Usability

1. Core Value: Decision Support

- Intuitive AI Insights: `Analytics` page converts complex database metrics (sales, profit, conversion rates) into Strategic Actions and LLM Reasoning. This is highly effective for Small-to-Medium Enterprise (SME) owners who may not understand raw data but can easily act on advice like "Restock immediately" or "Increase marketing exposure."
- CEO Report Feature: The one-click generation of a store-wide summary (CEO Report) significantly reduces "information overload" for the user.

2. Interaction Experience

- Real-time Feedback (Syncing Logic):

1. Strength: Implemented a "Sync to Database" logic that ensures data consistency.
2. Weakness: The current Dropdown refresh lag is a significant usability hurdle. If a user adds a product but cannot find it on the analysis page, they will perceive the app as "broken."

- Fix: Implement the `didChangeDependencies` auto-refresh or a global state update to ensure a seamless flow.
- Form Design: Using `Row` layouts for price and cost in the `ProductPage` is space-efficient for mobile screens and aligns with mobile UI best practices.

Demo of EZ System

Conclusion

EZ_System transforms raw e-commerce and economic data into clear, AI-driven insights that help SMEs make better business decisions. By combining internal metrics with CPI and PPI data through a hybrid intelligence engine, it generates actionable recommendations, trade-off analysis, and demand forecasts. This system reduces reliance on guesswork and empowers non-technical users to optimize pricing, inventory, and profitability more effectively.

Thank you!